

최신 시계열 모형을 활용한 온라인 판매량 예측

: ARIMAX, Prophet, DLinear 모형을 중심으로

시간의 지평선

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2. 데이터 소개

3. 분석 모델

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Part 1 서론



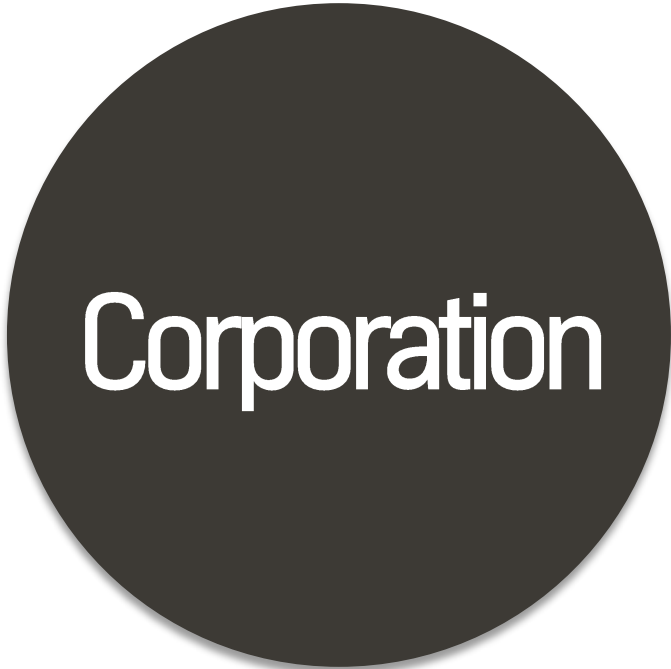
Data

실시간 축적 및 수집



Forecast

정확한 판매 내용 예측



Corporation

마케팅 전략 수립
비용이나 자원의 효율적 관리

Part 2 데이터 소개

데이터 수집 및 EDA

brand_keyword_cnt

브랜드 별 일별 언급량 수치의 정규화 데이터

Product_info

제품 별 제품특성 및 제원 소개 데이터

sales

실제 판매된 제품들의 명세와 일별 판매금액 (단위: 원)

train

실제 판매된 제품들의 명세와 일별 판매량 (단위: 개수)

EDA

[Exploratory Data Analysis, 탐색적 데이터분석]

”

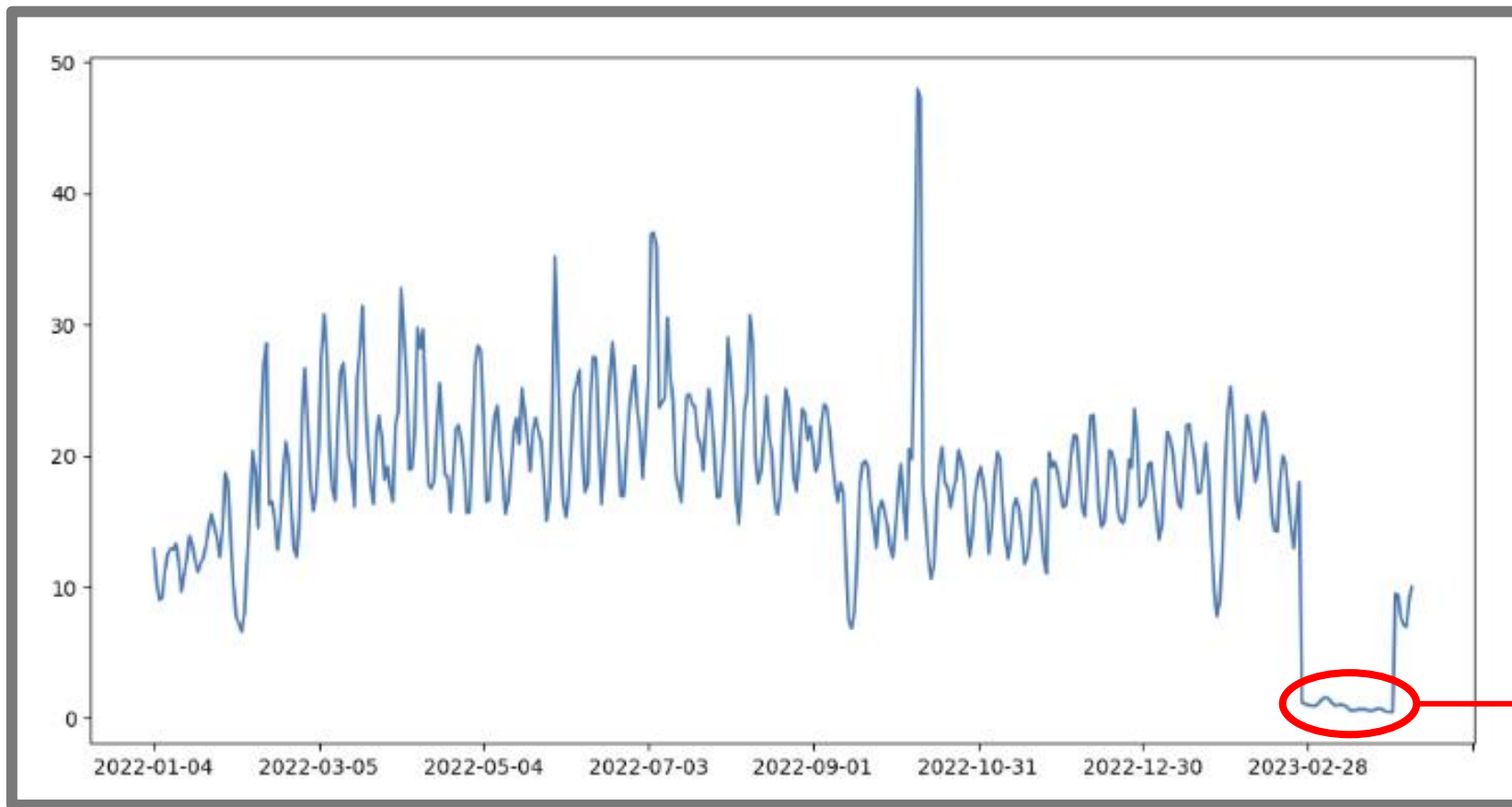
다양한 통계적 기법과

데이터 시각화 기법 등을 활용하여

데이터를 분석, 조사하여

주요 특징을 요약하는 과정

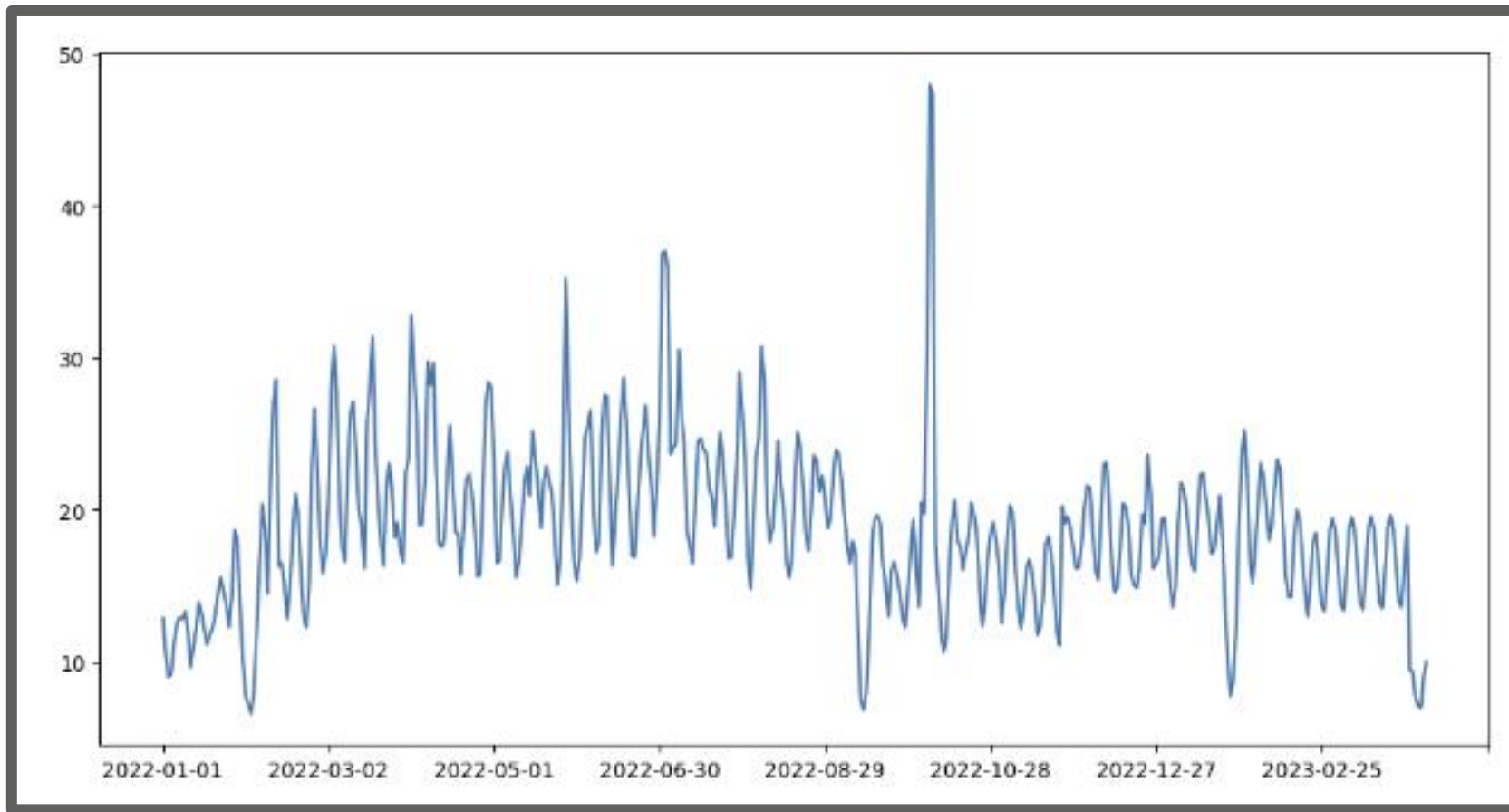
Part 2 EDA 및 전처리



2023년 2월 23일부터
3월 28일까지의
판매량 데이터

→ 보간(Interpolation)
필요

Part 2 EDA 및 전처리



◀ Prophet 모델
보간한 그래프

Part 3 분석 모델

ARIMAX, Prophet, DLinear

ARIMA

AR(자기회귀)모형

$$Y_t = c + \underbrace{\phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p}}_{\text{AR(자기회귀)모형}} + \underbrace{\theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \dots + \theta_q \varepsilon_{t-q} + \varepsilon_t}_{\text{MA(이동평균)모형}}$$

MA(이동평균)모형

ARIMAX

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \dots + \theta_q \varepsilon_{t-q} + \underbrace{\beta_1 X_{1,t} + \beta_2 X_{2,t} + \dots + \beta_k X_{k,t}}_{\text{기존모형에 독립변수 추가}} + \varepsilon_t$$

기존모형에 독립변수 추가

Time series Forecast 방법론

전통적인 통계학 기반 방법론 : ARIMA, SARIMA, ...

머신 러닝 기법 : Gradient Boosting Regression Tree

딥러닝 기법

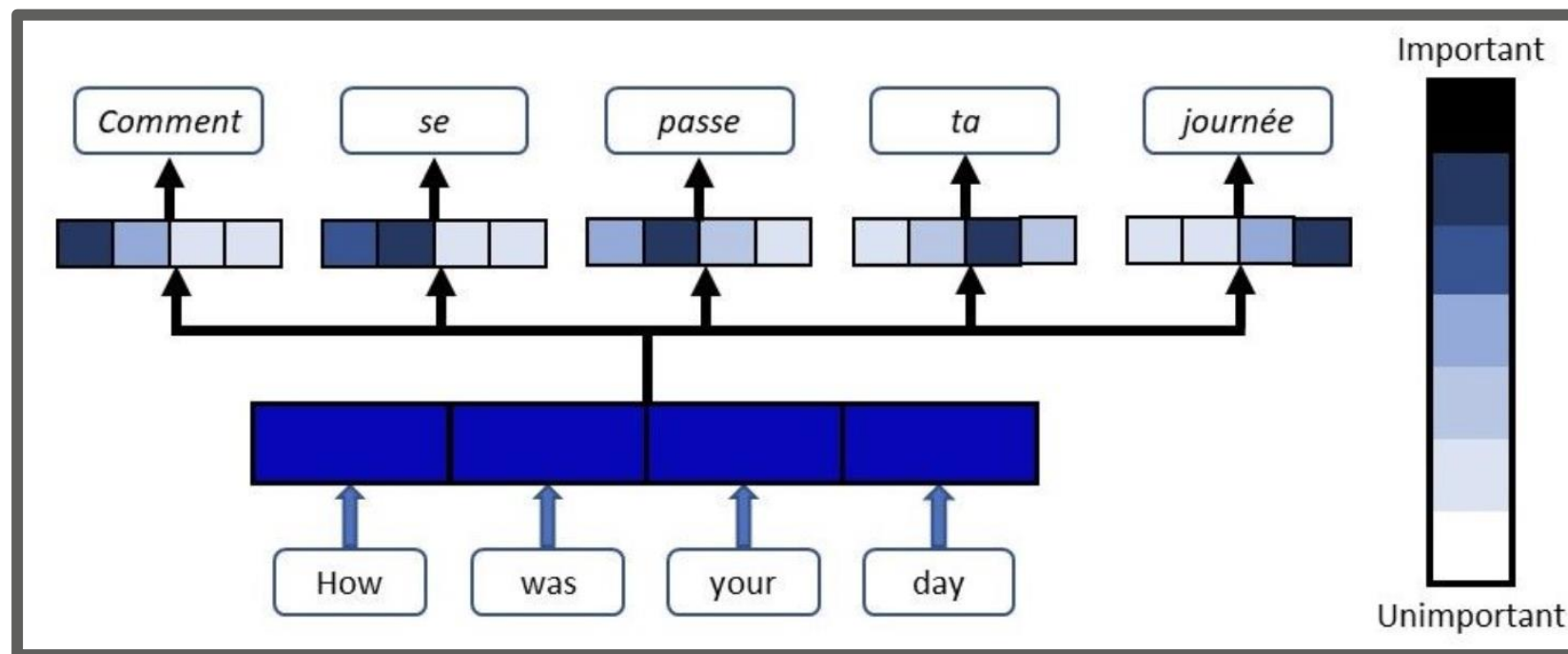
RNN 기반 모델 : RNN, LSTM, GRU

CNN 기반 모델 : TCN

Transformer 기반 모델 : Informer, FEDformer

Autoregressive 방식으로 예측 -> 잠재적 Error Accumulation Effect

Are Transformers Effective for Time Series Forecasting? (zeng et al., 2022)



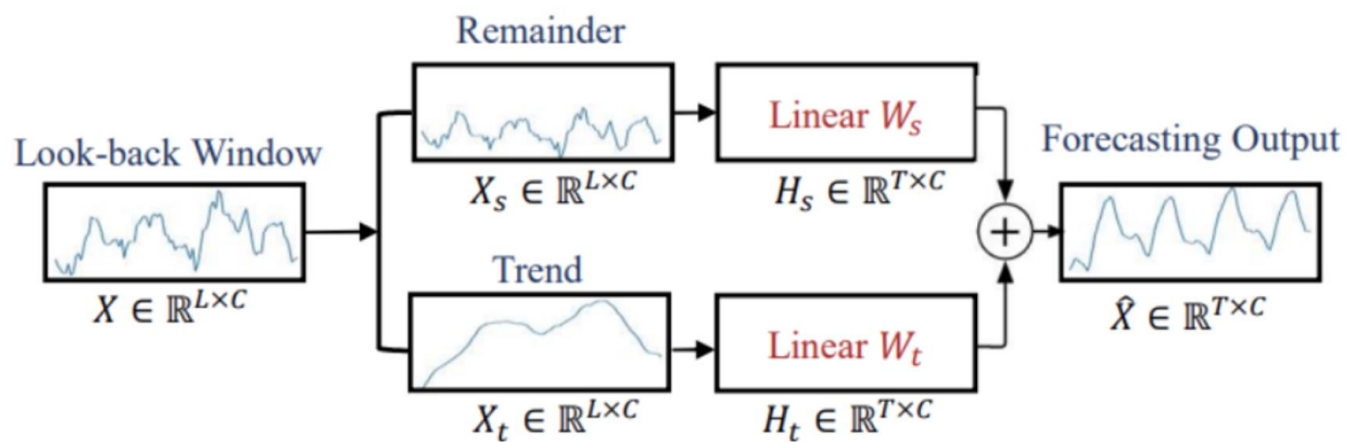
Semantic Correlation between Paired Element

나는 빨간색 사과를 먹는다.

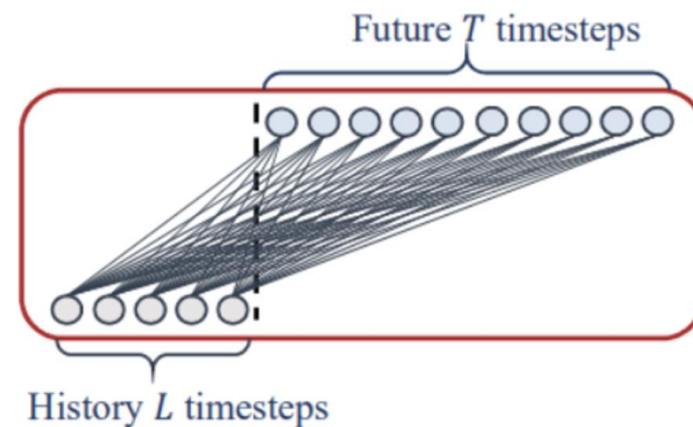
The diagram illustrates semantic correlation between the words '빨간색' (red) and '사과' (apple) in the sentence '나는 빨간색 사과를 먹는다.' (I eat red apples). A dashed rectangular box is drawn above the words '빨간색' and '사과', with solid lines connecting the top of each word to the top of the box, indicating a semantic relationship between them.

▶ Temporal Information Loss 발생

Part 3 DLinear



(a) The whole structure of *DLinear*



(b) One Linear Layer

$$\text{Remainder} : H_s = W_s X_s$$

$$\hat{X} = H_s + H_t$$

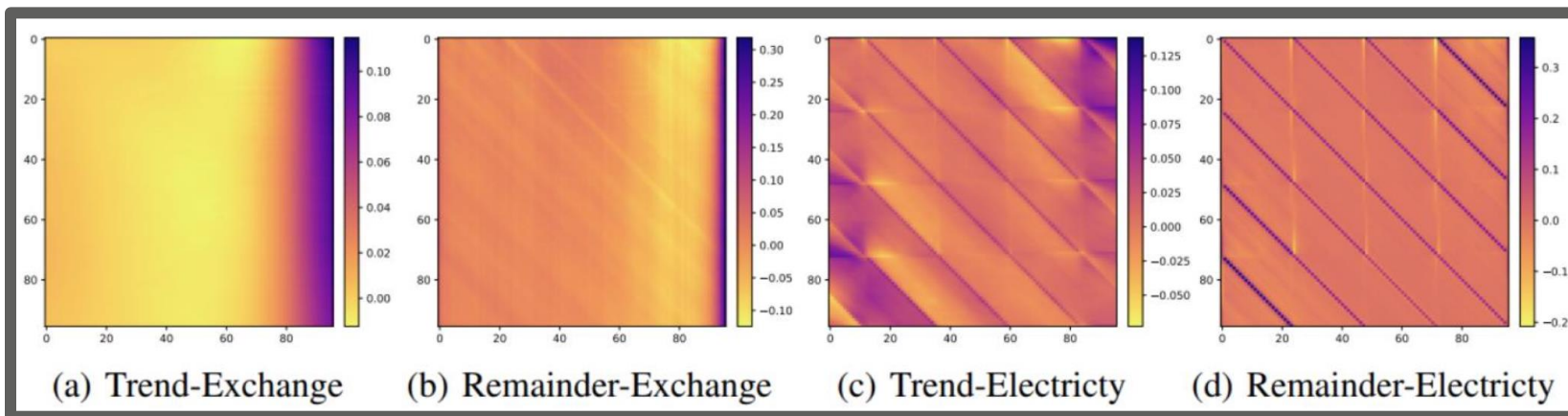
$$\text{Trend} : H_t = W_t X_t$$

Part 3 DLinear

Time Series Benchmark

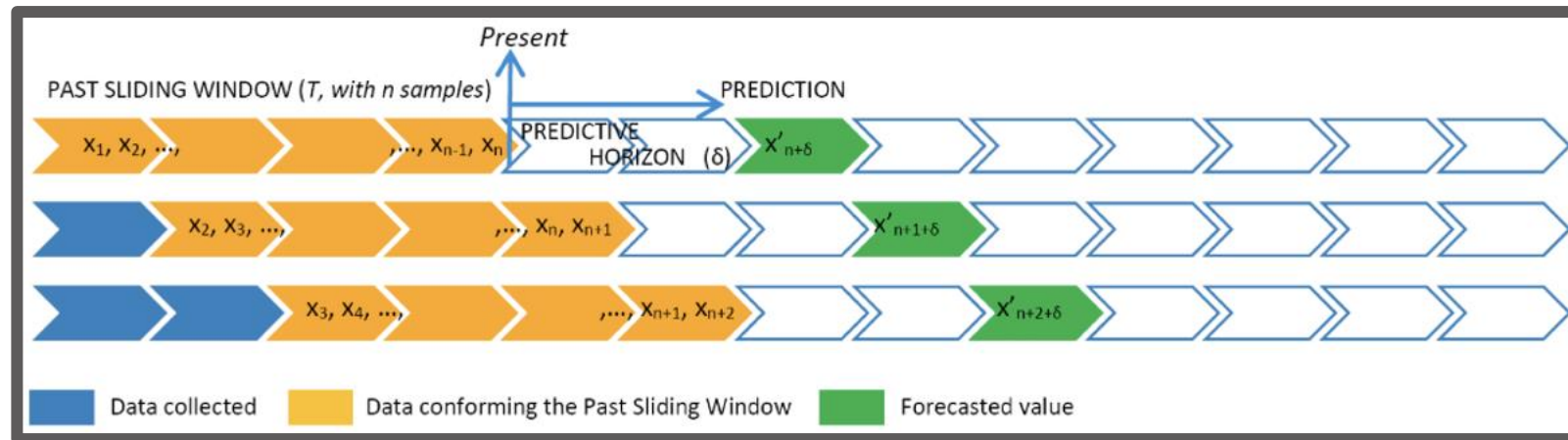
Methods	Metric	Electricity				Exchange-Rate				Traffic				Weather				ILI			
		96	192	336	720	96	192	336	720	96	192	336	720	96	192	336	720	24	36	48	60
<i>DLinear-S*</i>	MSE	0.194	<u>0.193</u>	<u>0.206</u>	<u>0.242</u>	0.078	<u>0.159</u>	<u>0.274</u>	0.558	0.650	0.598	0.605	<u>0.645</u>	<u>0.196</u>	<u>0.237</u>	<u>0.283</u>	<u>0.345</u>	2.398	2.646	<u>2.614</u>	<u>2.804</u>
	MAE	<u>0.276</u>	<u>0.280</u>	<u>0.296</u>	<u>0.329</u>	<u>0.197</u>	<u>0.292</u>	<u>0.391</u>	0.574	0.396	0.370	<u>0.373</u>	<u>0.394</u>	0.255	0.296	<u>0.335</u>	<u>0.381</u>	1.040	1.088	1.086	1.146
<i>DLinear-I*</i>	MSE	0.184	0.184	0.197	0.234	0.084	0.157	0.236	<u>0.626</u>	0.647	<u>0.602</u>	<u>0.607</u>	0.646	0.164	0.209	0.263	0.338	<u>3.015</u>	2.737	2.577	2.821
	MAE	0.270	0.273	0.289	0.323	0.216	0.298	0.379	<u>0.634</u>	0.403	0.375	0.377	0.398	0.237	0.282	0.327	0.380	<u>1.192</u>	1.036	<u>1.043</u>	1.091
FEDformer	MSE	<u>0.193</u>	0.201	0.214	0.246	0.148	0.271	0.460	1.195	0.587	0.604	0.621	0.626	0.217	0.276	0.339	0.403	3.228	<u>2.679</u>	2.622	2.857
	MAE	<u>0.308</u>	0.315	0.329	0.355	0.278	0.380	0.500	0.841	0.366	<u>0.373</u>	0.383	0.382	0.296	0.336	0.380	0.428	1.260	<u>1.080</u>	<u>1.078</u>	1.157
Autoformer	MSE	0.201	0.222	0.231	0.254	0.197	0.300	0.509	1.447	<u>0.613</u>	0.616	0.622	0.660	0.266	0.307	0.359	0.419	3.483	3.103	2.669	2.770
	MAE	0.317	0.334	0.338	0.361	0.323	0.369	0.524	0.941	0.388	0.382	0.337	0.408	0.336	0.367	0.395	0.428	1.287	1.148	1.085	<u>1.125</u>
Informer	MSE	0.274	0.296	0.300	0.373	0.847	1.204	1.672	2.478	0.719	0.696	0.777	0.864	0.300	0.598	0.578	1.059	5.764	4.755	4.763	5.264
	MAE	0.368	0.386	0.394	0.439	0.752	0.895	1.036	1.310	0.391	0.379	0.420	0.472	0.384	0.544	0.523	0.741	1.677	1.467	1.469	1.564
Pyraformer*	MSE	0.386	0.378	0.376	0.376	1.748	1.874	1.943	2.085	0.867	0.869	0.881	0.896	0.622	0.739	1.004	1.420	7.394	7.551	7.662	7.931
	MAE	0.449	0.443	0.443	0.445	1.105	1.151	1.172	1.206	0.468	0.467	0.469	0.473	0.556	0.624	0.753	0.934	2.012	2.031	2.057	2.100
LogTrans	MSE	0.258	0.266	0.280	0.283	0.968	1.040	1.659	1.941	0.684	0.685	0.734	0.717	0.458	0.658	0.797	0.869	4.480	4.799	4.800	5.278
	MAE	0.357	0.368	0.380	0.376	0.812	0.851	1.081	1.127	<u>0.384</u>	0.390	0.408	0.396	0.490	0.589	0.652	0.675	1.444	1.467	1.468	1.560
Reformer	MSE	0.312	0.348	0.350	0.340	1.065	1.188	1.357	1.510	0.732	0.733	0.742	0.755	0.689	0.752	0.639	1.130	4.400	4.783	4.832	4.882
	MAE	0.402	0.433	0.433	0.420	0.829	0.906	0.976	1.016	0.423	0.420	0.420	0.423	0.596	0.638	0.596	0.792	1.382	1.448	1.465	1.483
Repeat-C*	MSE	1.588	1.595	1.617	1.647	<u>0.081</u>	0.167	0.305	0.823	2.723	2.756	2.791	2.811	0.259	0.309	0.377	0.465	6.587	7.130	6.575	5.893
	MAE	0.946	0.950	0.961	0.975	0.196	0.289	0.396	0.681	1.079	1.087	1.095	1.097	<u>0.254</u>	<u>0.292</u>	0.338	0.394	1.701	1.884	1.798	1.677

Part 3 DLinear



Method	MACs	Parameter	Time	Memory	Time	Memory	Test Step
<i>DLinear</i>	0.04G	139.7K	0.4ms	687MiB	$O(L)$	$O(L)$	1
Transformer $\times$	4.03G	13.61M	26.8ms	6091MiB	$O(L^2)$	$O(L^2)$	1
Informer	3.93G	14.39M	49.3ms	3869MiB	$O(L\log L)$	$O(L\log L)$	1
Autoformer	4.41G	14.91M	164.1ms	7607MiB	$O(L\log L)$	$O(L\log L)$	1
Pyraformer	0.80G	241.4M*	3.4ms	7017MiB	$O(L)$	$O(L)$	1
FEDformer	4.41G	20.68M	40.5ms	4143MiB	$O(L)$	$O(L)$	1

Time Slide



Look back window : 90 Day
Predict window : 21 Day

Prophet : 'Facebook(현 Meta)'에서 개발한 시계열 모델



추세(trend), 계절성(seasonality),
불규칙하거나 위의 두 성질로는 설명되지 않는
임의성(randomness) 등의 성분이 포함된
시계열 데이터를 개별 성분, 요인으로 분해하여
변화의 특성을 파악할 수 있도록 함

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t.$$

추세

공휴일

계절성

시간에 따라 변화하도록 time 고려

$$g(t) = \frac{C(t)}{1 + \exp(-(k + \mathbf{a}(t)^\top \boldsymbol{\delta})(t - (m + \mathbf{a}(t)^\top \boldsymbol{\gamma})))}$$

성장률의 Change Point를 고려하도록 변경

$$s(t) = \sum_{n=1}^N \left(a_n \cos \left(\frac{2\pi n t}{P} \right) + b_n \sin \left(\frac{2\pi n t}{P} \right) \right)$$

Fourier order

Fourier Series 계수

Fourier Series 계수

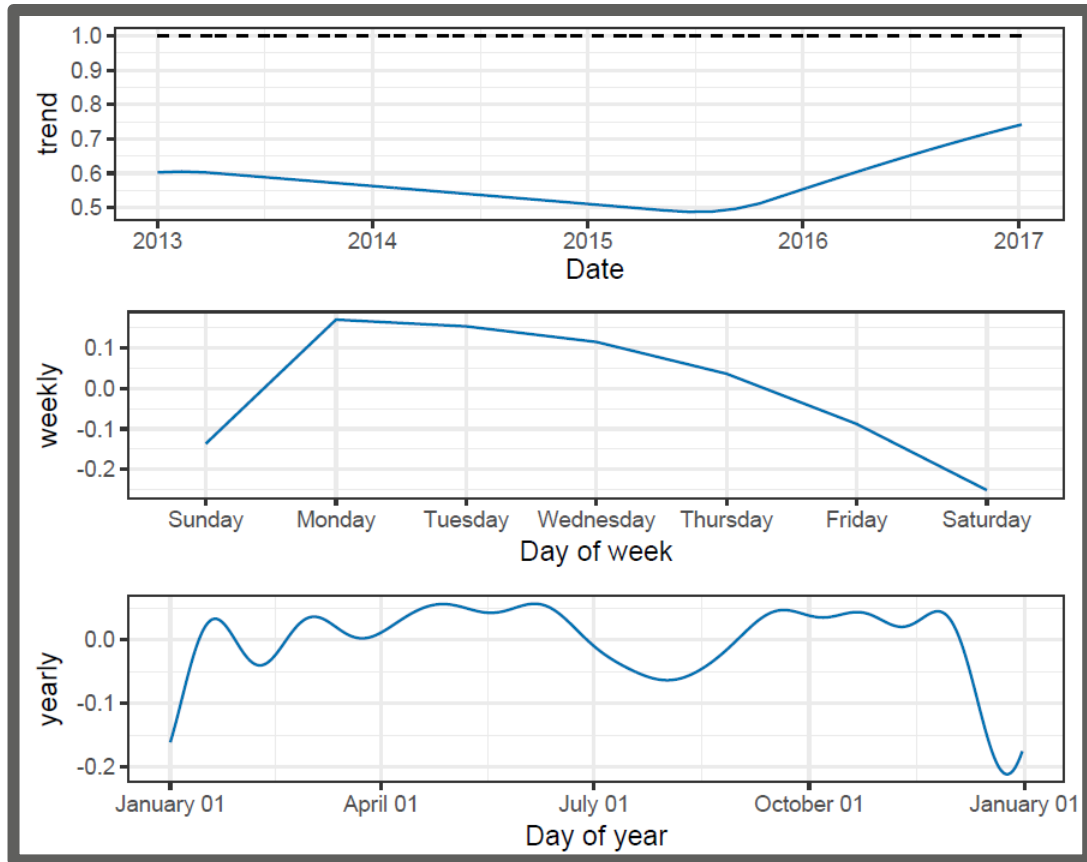
계절성 주기

각각의 휴일

$$Z(t) = [1(t \in D_1), \dots, 1(t \in D_L)]$$

$$h(t) = Z(t)\kappa.$$

Part 3 Prophet



예측에 활용된 각 구성요소에 대한
확인 가능성이
Decompose Model의 이점



각 component가 어떻게 영향을
미치는지 그래프를 통해 확인 가능
이를 통해 예측 뿐만 아니라
추가적인 Insight 도출도 가능

Part 4 결론

Part 4 Pseudo SFA Score 비교

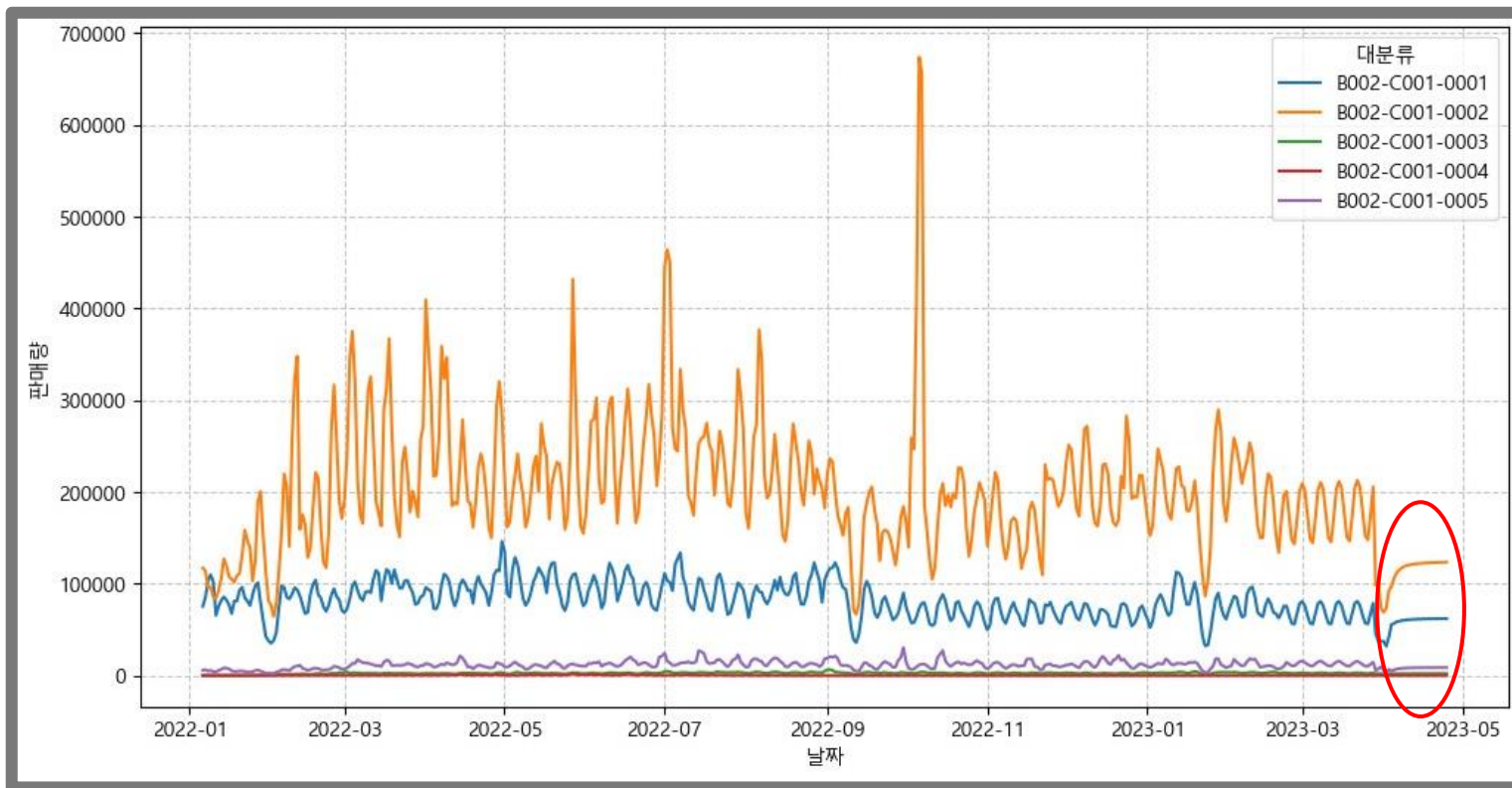
대분류 별 Pseudo 예측 정확도: $PSFA_m = 1 - \frac{1}{n} \sum_{day=1}^n \sum_{i=1}^N \left(\frac{|y_i^{day} - p_i^{day}|}{\max(y_i^{day}, p_i^{day})} \right) \times \frac{y_i^{day}}{\sum_{i=1}^N y_i^{day}}$

오차 (판매)비중

- m : 대분류 index
- i : (대분류 내에서) 제품 index
- y_i^{day} : i 번째 제품의 day 일의 판매량
- p_i^{day} : i 번째 제품의 day 일의 예측량

전체 Pseudo 예측 정확도: $PSFA = \frac{1}{M} \sum_{m=1}^M PSFA_m$

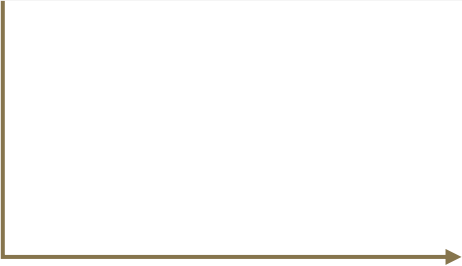
Part 4 분석 결과: ARIMA



Pseudo-SFA score: 0.479

키워드 언급량 데이터 예측

브랜 드	2022-01- 01	2022-01- 02	2022-01- 03	2022-01- 04	2022-01- 05	2022-01- 06	2022-01- 07	2022-01- 08	2022-01- 09	...	2023-03- 26	2023-03- 27	2023-03- 28	2023-03- 29
0 B002- 00001	0.84131	0.91383	1.450530	2.422390	1.871190	1.581080	1.232950	1.174930	1.145920	...	0.319110	0.391640	0.377130	0.49318
1 B002- 00002	12.64868	20.27850	15.332170	12.750210	13.562510	13.707570	11.937910	15.564250	14.084710	...	10.269790	11.966920	10.646930	10.41485
2 B002- 00003	0.33362	0.43516	0.362630	0.174060	0.217580	0.464170	0.420650	0.290100	0.377130	...	0.536690	0.696250	0.449660	0.39164
3 B002- 00005	1.07339	1.71163	2.016240	1.914700	1.987230	2.146790	1.682620	1.378000	1.421520	...	2.219320	2.509420	2.872060	2.37888
4 B002- 00006	0.00000	0.00000	0.188558	0.246574	0.246574	0.246574	0.377139	0.087012	0.261084	...	0.072526	0.290103	0.087012	0.00000
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
3165 B002- 03794	2.32085	2.98810	3.611830	4.061500	3.669850	3.771390	3.031620	2.988100	3.133150	...	2.422390	2.422390	2.756010	2.32085
3166 B002- 03795	0.14505	0.00000	0.087030	0.072520	0.087030	0.101530	0.072520	0.130540	0.116040	...	0.000000	0.072520	0.000000	0.10153
3167 B002- 03796	0.00000	0.00000	0.000000	0.000000	0.000000	0.101530	0.000000	0.000000	0.000000	...	0.000000	0.000000	0.000000	0.00000
3168 B002- 03798	0.14505	0.00000	0.116040	0.072520	0.116040	0.275600	0.217580	0.116040	0.101530	...	0.101530	0.087030	0.145050	0.17406
3169 B002- 03799	4.55468	5.54105	6.150270	6.396860	7.006090	6.657960	5.802140	5.483020	6.034230	...	5.105880	6.672460	6.440380	5.90368



각 제품 별로 매칭(merge)

이후 ARIMA를 통해 4월5일부터 25일까지에 대한 예측 진행

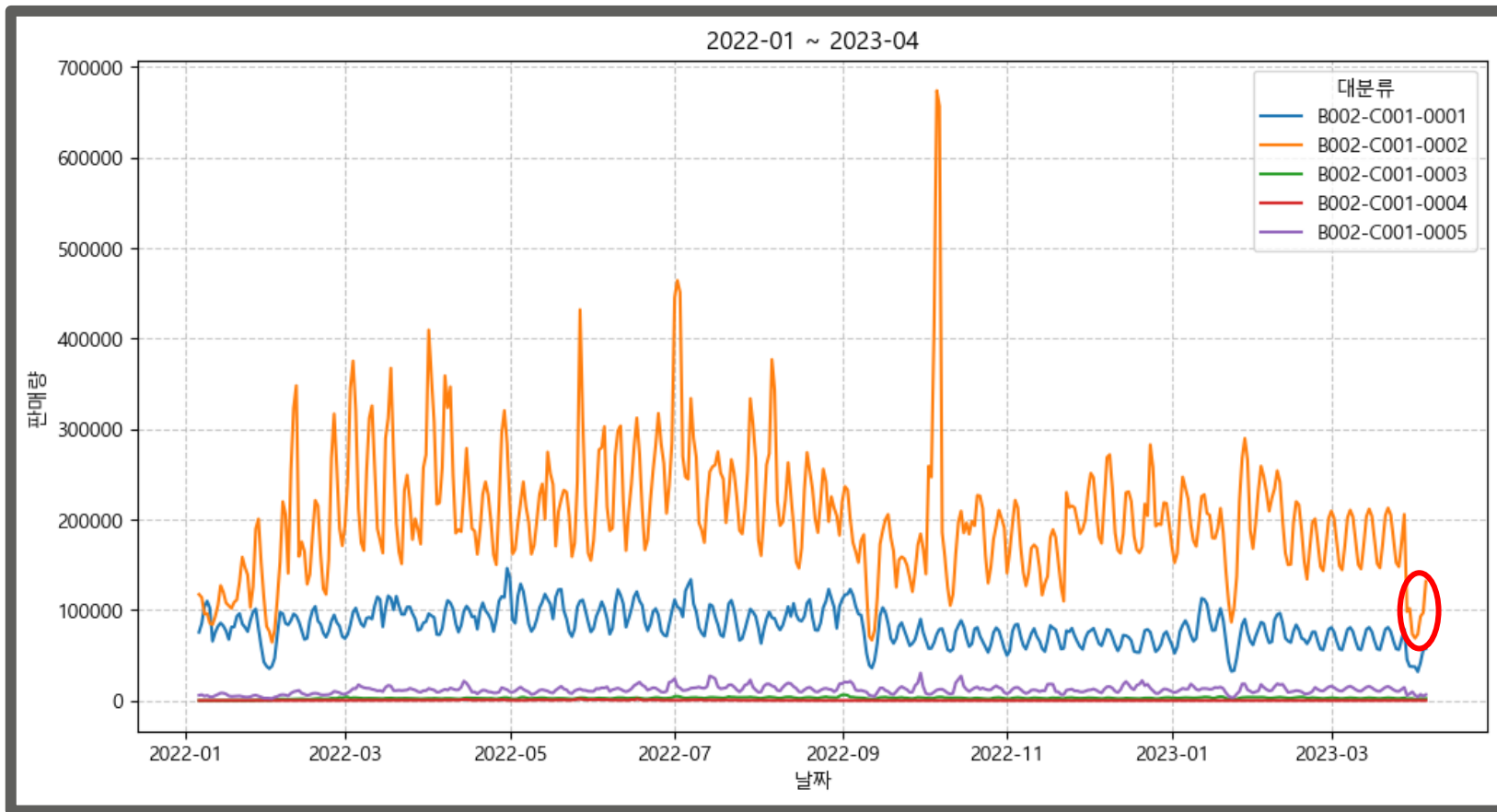
	ID	2022-01- 01	2022-01- 02	2022-01- 03	2022-01- 04	2022-01- 05	2022-01- 06	2022-01- 07	2022-01- 08	2022-01- 09	...	2023-04- 16	2023-04- 17	2023-04- 18	2023-04- 19
0	0	0.84131	0.91383	1.45053	2.42239	1.87119	1.58108	1.23295	1.17493	1.14592	...	0.371166	0.371166	0.371166	0.371166
1	1	12.64868	20.27850	15.33217	12.75021	13.56251	13.70757	11.93791	15.56425	14.08471	...	10.506437	10.506432	10.506430	10.506429
2	2	12.64868	20.27850	15.33217	12.75021	13.56251	13.70757	11.93791	15.56425	14.08471	...	10.506437	10.506432	10.506430	10.506429
3	3	12.64868	20.27850	15.33217	12.75021	13.56251	13.70757	11.93791	15.56425	14.08471	...	10.506437	10.506432	10.506430	10.506429
4	4	0.33362	0.43516	0.36263	0.17406	0.21758	0.46417	0.42065	0.29010	0.37713	...	0.745183	0.745183	0.745183	0.745183
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
15885	15885	4.55468	5.54105	6.15027	6.39686	7.00609	6.65796	5.80214	5.48302	6.03423	...	5.116046	5.116047	5.116047	5.116047
15886	15886	4.55468	5.54105	6.15027	6.39686	7.00609	6.65796	5.80214	5.48302	6.03423	...	5.116046	5.116047	5.116047	5.116047
15887	15887	4.55468	5.54105	6.15027	6.39686	7.00609	6.65796	5.80214	5.48302	6.03423	...	5.116046	5.116047	5.116047	5.116047
15888	15888	4.55468	5.54105	6.15027	6.39686	7.00609	6.65796	5.80214	5.48302	6.03423	...	5.116046	5.116047	5.116047	5.116047
15889	15889	4.55468	5.54105	6.15027	6.39686	7.00609	6.65796	5.80214	5.48302	6.03423	...	5.116046	5.116047	5.116047	5.116047

auto.arima로 차수 결정

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \dots + \theta_q \epsilon_{t-q} + \underbrace{\beta_1 X_{1,t} + \beta_2 X_{2,t} + \dots + \beta_k X_{k,t}}_{\text{브랜드 키워드 언급량}} + \epsilon_t$$

브랜드 키워드 언급량

Part 4 분석 결과: ARIMAX



Pseudo-SFA score: 0.491

Part 4 분석결과: ARIMAX

ARIMA

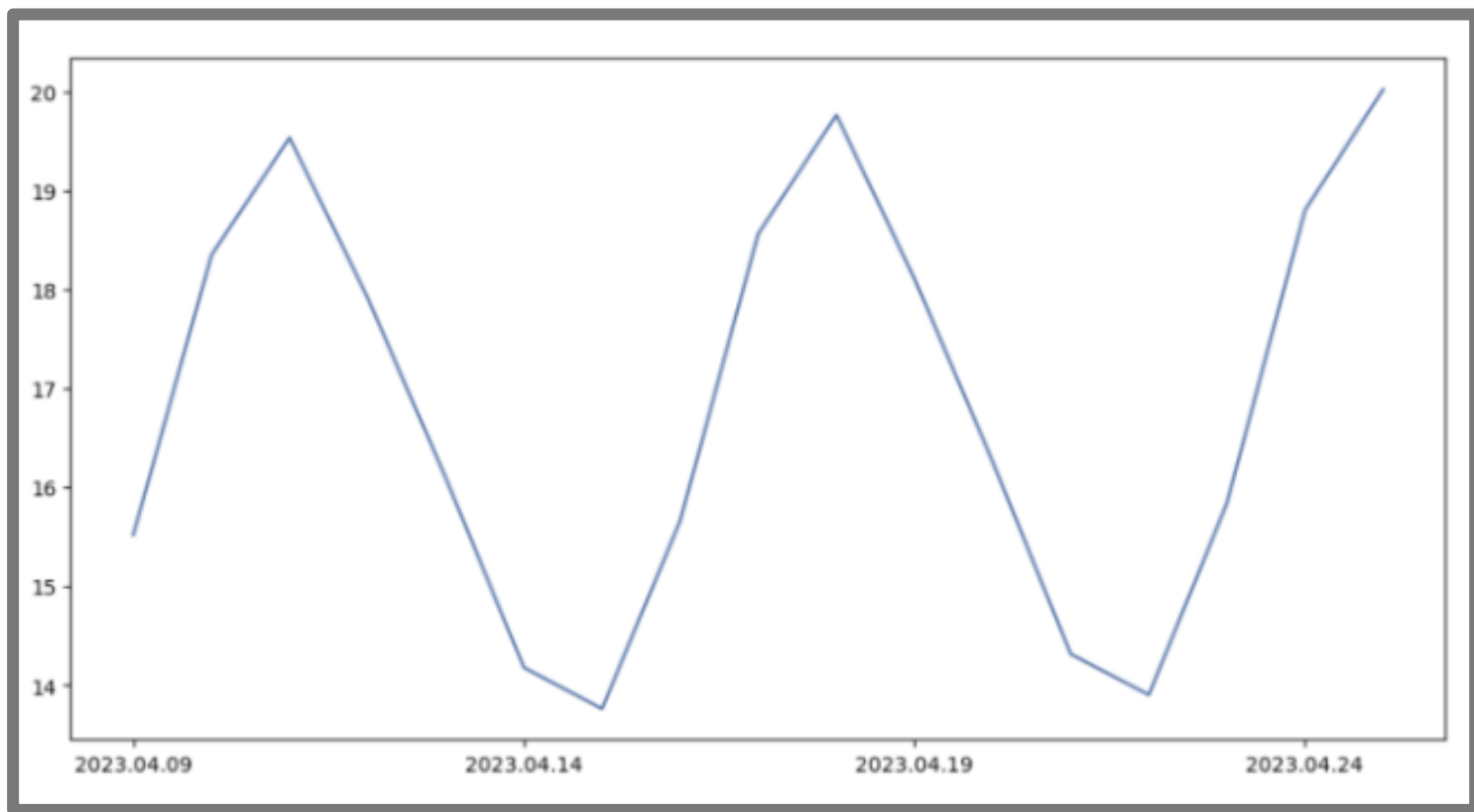
대분류	B002-C001-0001	B002-C001-0002	B002-C001-0003	B002-C001-0004	B002-C001-0005
2023-04-05	56596.521642	106203.091173	1956.889157	121.142638	6260.125144
2023-04-06	58731.559274	111922.381528	1990.518241	118.962810	7082.869708
2023-04-07	59029.589693	115202.614814	1993.121026	117.819932	7590.573047
2023-04-08	60083.910627	117462.452419	2022.060714	117.310993	7898.025150
2023-04-09	60100.321366	118929.491763	2020.591591	116.769363	8118.150113
2023-04-10	60846.309246	119937.919136	2044.478190	116.628464	8256.692041
2023-04-11	60698.267710	120690.315542	2040.997949	116.377237	8367.525213
2023-04-12	61298.881367	121173.859461	2061.659131	116.383022	8435.344368
2023-04-13	61070.271284	121622.185341	2057.453102	116.275812	8496.125874
2023-04-14	61582.351888	121866.401323	2076.154633	116.334481	8530.313875
2023-04-15	61317.423032	122181.393505	2071.778199	116.295342	8565.740959
2023-04-16	61768.709931	122308.752451	2089.260885	116.366461	8582.824615
2023-04-17	61490.813142	122565.006337	2084.940715	116.358494	8604.604392
2023-04-18	61897.295147	122631.491657	2101.623629	116.426023	8612.653039
2023-04-19	61619.096751	122862.552217	2097.446483	116.431784	8626.766396
2023-04-20	61990.759167	122895.196450	2113.569621	116.490452	8629.937991
2023-04-21	61719.162686	123116.415637	2109.563020	116.501794	8639.593754
2023-04-22	62062.557671	123128.926921	2125.268255	116.550817	8640.106771
2023-04-23	61801.217311	123347.300313	2121.434974	116.563973	8647.090132
2023-04-24	62120.828583	123347.022349	2136.809562	116.604247	8646.156994
2023-04-25	61871.523732	123565.647954	2133.142600	116.617504	8651.490926

ARIMAX - 더 높은 성능을 보이거나 판매량에 대한 Overestimation

대분류	B002-C001-0001	B002-C001-0002	B002-C001-0003	B002-C001-0004	B002-C001-0005
2023-04-05	5.741215e+04	1.317245e+05	2.087513e+03	114.376584	6434.264393
2023-04-06	5.999097e+04	3.376714e+08	2.287219e+03	108.065778	7348.659432
2023-04-07	1.037899e+10	6.524504e+12	2.750172e+03	103.919852	7934.984195
2023-04-08	8.790673e+04	1.175932e+17	4.012843e+03	101.388154	8291.174851
2023-04-09	1.082614e+18	2.247547e+21	7.760629e+03	99.383640	8549.785783
2023-04-10	1.173273e+06	4.096638e+25	1.925385e+04	98.248468	8715.638850
2023-04-11	1.129262e+26	7.827328e+29	5.528113e+04	97.266319	8847.603275
2023-04-12	6.610053e+09	1.427161e+34	1.702266e+05	96.768165	8931.690393
2023-04-13	1.177920e+34	2.730185e+38	5.426105e+05	96.282823	9005.904758
2023-04-14	1.614678e+15	4.971853e+42	1.765016e+06	96.074097	9050.542998
2023-04-15	1.228675e+42	9.525668e+46	5.822708e+06	95.829802	9095.925982
2023-04-16	3.971143e+20	1.732063e+51	1.941683e+07	95.749978	9120.227772
2023-04-17	1.281616e+50	3.323930e+55	6.530311e+07	95.623703	9150.357903
2023-04-18	9.766650e+25	6.034050e+59	2.211244e+08	95.599387	9163.667998
2023-04-19	1.336839e+58	1.159989e+64	7.527895e+08	95.531888	9185.363150
2023-04-20	2.402015e+31	<u>2.102104e+68</u>	2.573624e+09	95.530033	9192.479033
2023-04-21	1.394442e+66	<u>4.048581e+72</u>	8.827715e+09	95.492518	9209.326400
2023-04-22	5.907530e+36	<u>7.323178e+76</u>	3.035719e+10	95.498608	9212.799864
2023-04-23	1.454526e+74	<u>1.413193e+81</u>	1.045999e+11	95.476865	9226.759789
2023-04-24	1.452901e+42	<u>2.551202e+85</u>	3.609601e+11	95.484799	9227.988219
2023-04-25	1.517199e+82	<u>4.933471e+89</u>	1.247072e+12	95.471665	9240.169527

Part 4 분석결과: Prophet

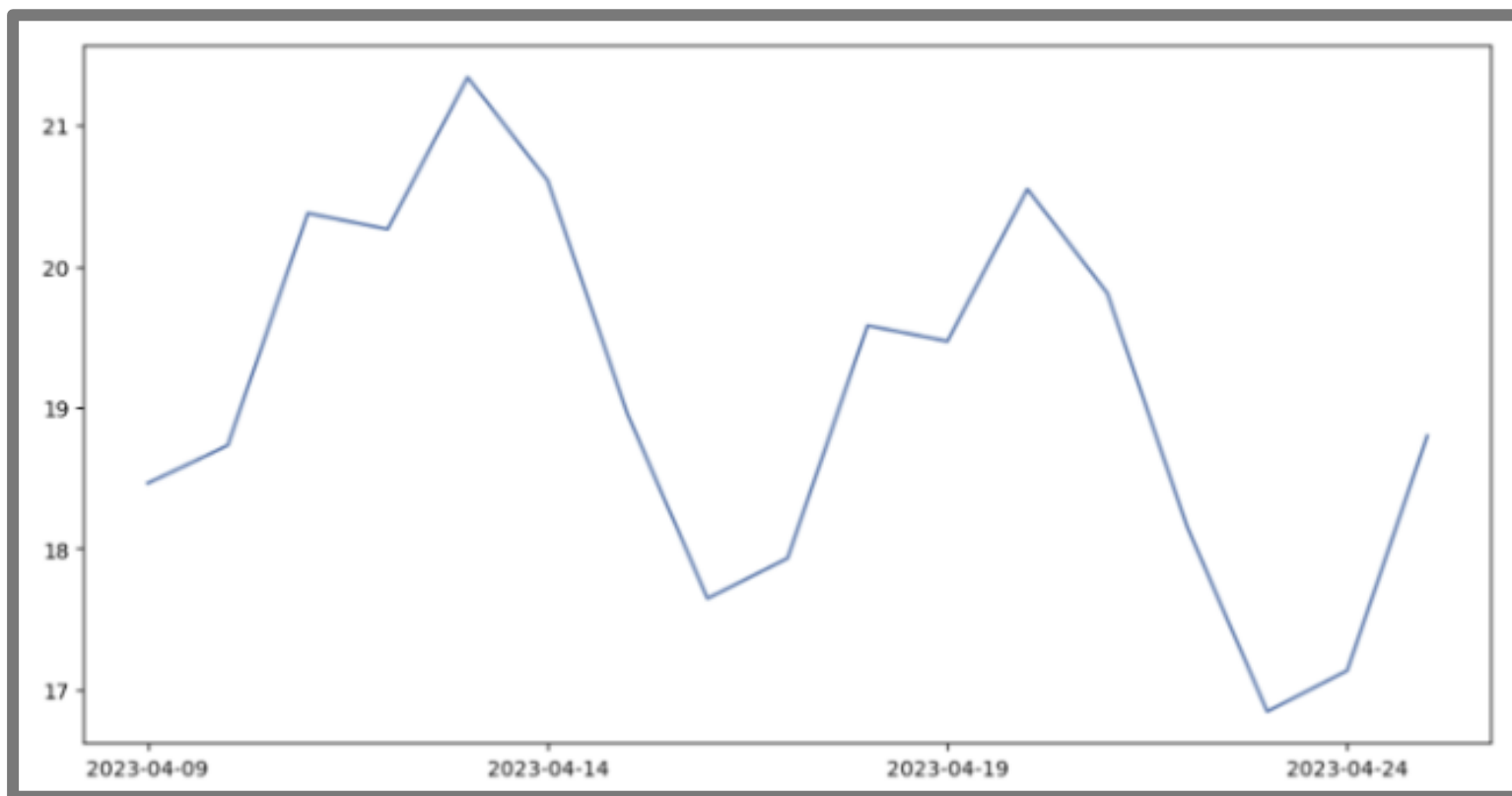
1. 2022-01-01 ~ 2023-04-04



Pseudo-SFA score: 0.246

Part 4 분석결과: Prophet

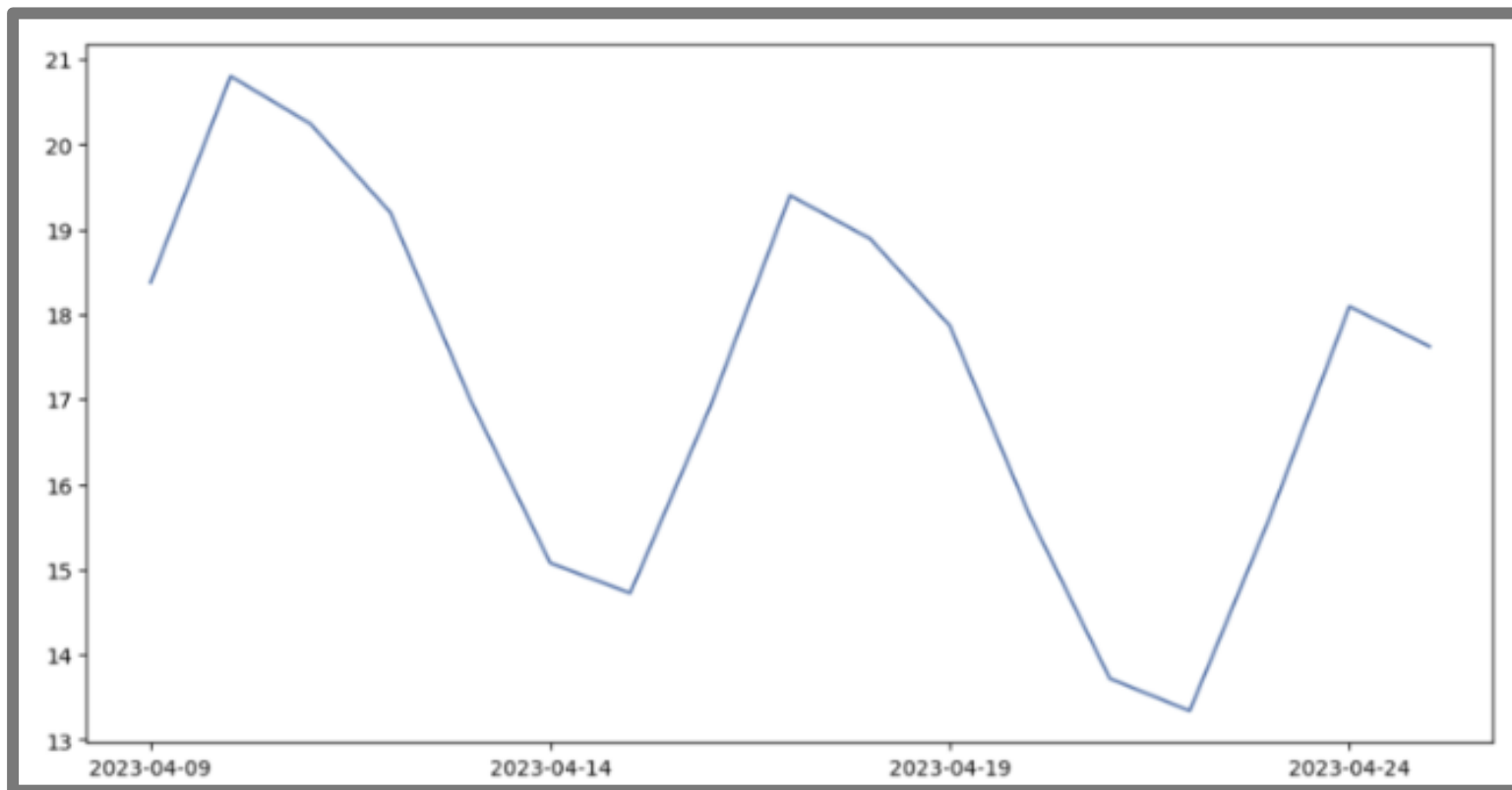
2. 2022-10-08 ~ 2023-04-04



Pseudo-SFA score: 0.349

Part 4 분석결과: Prophet

3. 2023-01-25 ~ 2023-04-04



Pseudo-SFA score: 0.432

분석 날짜	Pseudo-SFA score
2022-01-01 ~ 2023-04-04	0.24624
2022-10-08 ~ 2023-04-04	0.34979
2023-01-25 ~ 2023-04-04	0.43265

Part 4 분석 결과: DLinear

Model Compare

Layer (type)	Output Shape	Param #	Tr. Param #
LSTM-1	[1, 90, 512], [1, 1, 512], [1, 1, 512]	1,073,152	1,073,152
Linear-2	[1, 256]	131,328	131,328
ReLU-3	[1, 256]	0	0
Dropout-4	[1, 256]	0	0
Linear-5	[1, 21]	5,397	5,397
ReLU-6	[1, 21]	0	0
Total params: 1,209,877			
Trainable params: 1,209,877			
Non-trainable params: 0			

LSTM

Total Params : 1209877

Score : 0.49654

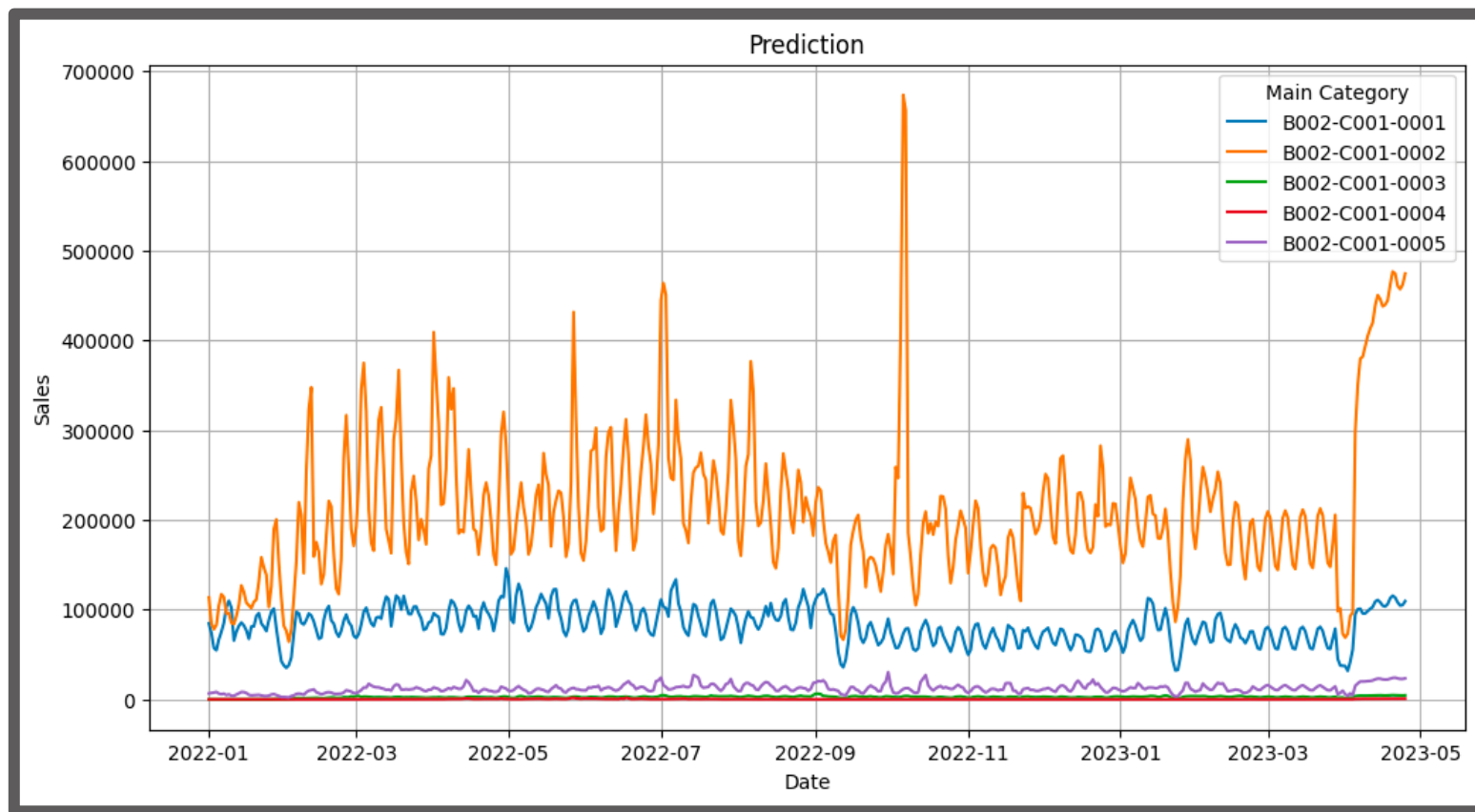
Layer (type)	Output Shape	Param #	Tr. Param #
series_decomp-1	[1, 90, 9], [1, 90, 9]	0	0
Linear-2	[1, 9, 21]	1,911	1,911
Linear-3	[1, 9, 21]	1,911	1,911
Total params: 3,822			
Trainable params: 3,822			
Non-trainable params: 0			

DLinear

Total Params : 3822

score: 0.568

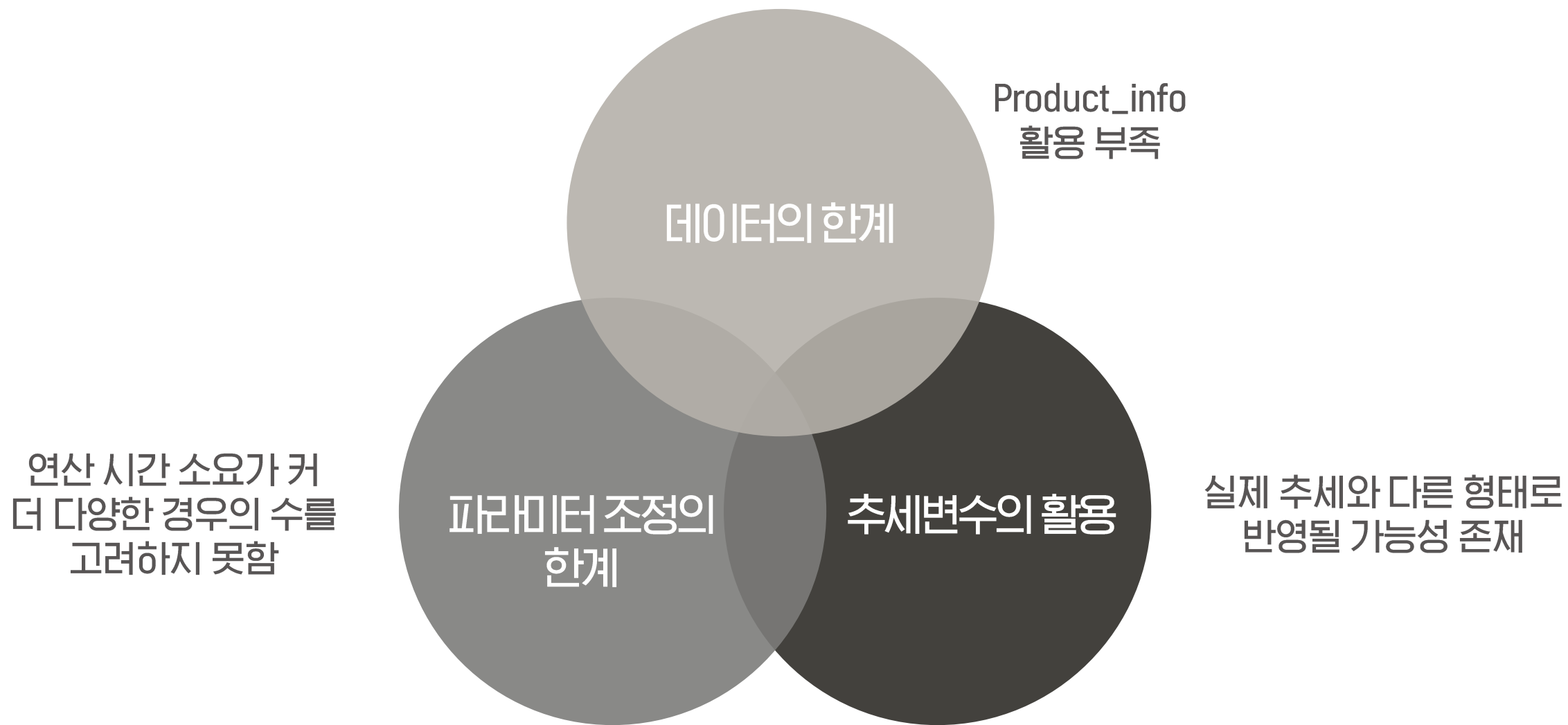
Part 4 분석 결과: DLinear



Pseudo-SFA score: 0.568



한계점 및 향후 연구 방향성



Q&A